

Risk Factor Engineering in Probability of Default Models

Tree-Based Interactions

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PD Model Development

- Probability of Default (PD) modeling consists of a few standardized steps. Although the process can differ significantly within specific steps, the overall modeling approach can be considered standardized, following certain principles.
- After collecting data, one of the first steps in PD modeling is to develop the risk differentiation function.
- Regardless of whether the model is used for regulatory or managerial purposes, this step is equally important and is often based on the same or similar principles.
- The main goal of the risk differentiation function is to ensure that the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- Typical steps in the development of the risk differentiation function include single-factor analysis, risk factor engineering, correlation analysis, and multi-factor analysis. These steps may overlap in certain areas and can, in essence, be considered integral parts of the risk differentiation function development, along with additional sub-steps within them.

PD Model Development cont.

- One of the most commonly used statistical methods in multi-factor analysis is logistic regression. This approach is typically considered the traditional method for PD model development.
- With the increasing use of machine learning (ML) in credit risk modeling, practitioners are exploring additional possibilities to enhance the development of the risk differentiation function using alternative methods.
- However, practitioners can still combine traditional and ML approaches while supporting the core principles of PD model development. One such example is risk factor engineering in PD modeling.
- The focus of this presentation is on risk factor engineering in PD modeling that goes beyond standard binning approaches. Specifically, the emphasis is on the derivation of new risk factors based on interactions identified by decision tree models. As the presented examples serve only for demonstration purposes, practitioners are encouraged to consider different use cases and designs for this approach.

Risk Factor Engineering in PD Modeling

- Risk factor engineering in PD modeling is one of the most important steps in the development of the risk differentiation function.
- In traditional PD model development, this step is often equated with the binning of risk factors (both numerical and categorical), although its application can extend far beyond this approach.
- Regardless of the final use of the PD models (regulatory or managerial), the principles followed in the binning step are largely shared.
- The following list presents some of the inputs and considerations typically used in a standard binning process:
 - minimum number of observations per bin;
 - minimum number of defaults (or target rate) per bin;
 - treatment of the missing or special values;
 - monotonic relationship between the risk factor and the target variable;
 - statistical or practical significance of risk factor bin profiles;
 - expert input and confirmation of the statistical relationship between the risk factor and the target variable.

Risk Factor Engineering in PD Modeling cont.

- When practitioners consider alternative risk factor engineering methods beyond binning, the same or similar principles are typically incorporated into this modeling step.
- The ultimate goal of these principles is to ensure full explainability of the impact and strong business support for the risk factors selected in the final model.
- One approach to risk factor engineering that goes beyond the standard binning process is tree-based interaction extraction. In essence, this method involves deriving new risk factors from decision tree models, which can then be used as variables in logistic regression or even as challenger models. When used as a challenger model, practitioners can also utilize the identified risk factors as supporting analysis in the risk factor selection process.
- The following slides present the main idea of a customized tree-based approach to risk factor engineering, along with a simulation study to demonstrate one potential use case.

Tree-Based Interactions

- Interactions between risk factors can be extracted from standard decision tree algorithms.
- However, standard algorithms typically do not provide all the inputs needed to support key considerations in risk factor engineering. For instance, requirements that are often unsupported or only partially supported include the minimum number of defaults or target rate, specific treatment of missing (and special) values, monotonicity constraints, statistical testing as a splitting criterion, and control over tree depth.
- Given the strong requirements for risk factor engineering in PD modeling, [Andrija Djurovic](#) proposed a customized tree-based approach. The main adjustments to the standard decision tree methodology relate to more flexible inputs:
 - minimum number of observations in the final nodes;
 - minimum target rate in the final nodes;
 - separate treatment of numerical and categorical risk factors;
 - maximum tree depth;
 - statistical testing as the splitting criterion;
 - monotonicity constraints at the splitting node;
 - treatment of missing values at the splitting node to ensure monotonicity;
 - overall flexibility for applying this approach across different modeling designs.
- The following slides present the basics of the splitting criteria, differences in the treatment of categorical and numerical risk factors, and a simulation study illustrating one possible use case of interactions as support for standard PD model development.

Tree-Based Interactions cont.

Splitting criterion - Gini Index

The Gini Index (or Gini impurity) is a standard splitting criterion used in decision tree algorithms for binary target variables. It measures the degree of class mixing within a dataset, where higher values indicate greater impurity. In this context, impurity reflects the extent to which both classes are present within a node; the more balanced the class composition, the higher the impurity. For the selected binary split, Gini Index is given as follows:

$$\text{Gini} = - \sum_{j \in \{\text{left}, \text{right}\}} \frac{n_j}{n} (1 - p_j^2)$$

where:

- n_j is the number of observations in node j (left or right);
- n is the total number of observations;
- p_j is the default rate in node j (left or right).

Adding a minus sign in the above formula turns the selection criterion into a maximization problem; thus, the split that maximizes the expression is selected for splitting at a given node.

Tree-Based Interactions cont.

Splitting criterion - Two-Sample Test of Proportions

The two-sample test of proportions can be used as an alternative splitting criterion. Specifically, the split with the smallest p-value is considered the best split. This test is commonly used in evaluating different aspects of PD models, as well as in binning processes.

The following formula presents the test statistic:

$$Z = \frac{p_1 - p_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where:

- p_1 and p_2 are the default rates in each node;
- n_1 and n_2 denote the number of observations in each node;
- p is the pooled proportion calculated as $\frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$;

Given the monotonicity constraints, this test can be applied as either a two-sided or a one-sided test for the purpose of the splitting criterion.

Tree-Based Interactions cont.

Treatment of Categorical and Numerical Risk Factors

In PD modeling, practitioners often work with a limited number of inherently categorical risk factors. For this reason, these variables are rarely fully automatically engineered and directly used in the modeling process. Even in tree-based methods, it is recommended to first examine and prepare categorical risk factors before further use. Accordingly, the customized decision tree process treats categorical risk factors using standard inputs such as the number of observations and target rate. However, it also applies an initial ordering based on the alphabetical sequence of risk factor modalities. This requires ensuring a meaningful ordering of categorical modalities.

On the other hand, numerical risk factors are initially split based on a predefined number of cut points derived from observed quantiles. Additional considerations, such as the number of observations and target rate, are also incorporated into this process. Monotonicity constraints are enforced at each splitting node. As a result, splitting points may be adjusted to preserve monotonicity, which is one of the key criteria in risk factor engineering used in credit risk. Practitioners should note that monotonicity at the splitting node does not necessarily ensure monotonicity throughout the full tree structure. However, it significantly increases the likelihood of achieving overall monotonic behavior.

For both types of risk factors, missing values are assigned to a node in a way that preserves the monotonicity constraint. As a result, missing value preprocessing in this sense is not needed.

In addition to these criteria, the customized algorithm can control overall tree depth as well as the p-value used in the two-sample test. This reduces the need for additional tree pruning, which is often required when other splitting metrics are used.

Simulation Study

The purpose of this simulation is to demonstrate one use case of tree-based interaction extraction as a support for standard PD model development. Specifically, three numerical risk factors (maturity, amount, age) are used as inputs to the customized decision tree algorithm. The output of this step is then used as an additional risk factor. The final PD model is specified in the form $\text{default} \sim \text{balance} + \text{rf.inter}$, where `rf.inter` denotes a variable extracted from the decision tree, with Weights of Evidence (WoE) encoding applied to the risk factor modalities.

The customized decision tree is run under the following setup:

- 1 the minimum percentage of observations in the final node is set to 10%;
- 2 the minimum percentage of defaults in the final node is set to 5%;
- 3 the monotonicity constraint in the splitting node is activated;
- 4 the maximum number of splitting points for all numerical variables is set to 50 at each splitting node;
- 5 the p-value from the two-sample test of proportions is used as the splitting criterion;
- 6 the p-value threshold is set at the 5% significance level, supporting the tree growing process.

For full replicability, the dataset used in this simulation is available at the following [link](#).

The following slides present the main results of the modeling simulation described above.

Simulation Results

Customized Decision Tree - Output Summary

```
## nobs filter ind
## 219 (`maturity`<=16 | is.na(`maturity`)) & (`age`<=34 & !is.na(`age`)) 1
## 224 (`maturity`<=16 | is.na(`maturity`)) & (`age`>34 | is.na(`age`)) 2
## 392 (`maturity`>16 & !is.na(`maturity`)) & `maturity`<=33 3
## 165 (`maturity`>16 & !is.na(`maturity`)) & `maturity`>33 4
```

Splitting Node Details

```
## rf.t rf method sm split y.l y.r m.n monotonicity
## numeric `maturity` p-value 0.0000 16 0.2122 0.3698 TRUE 1
## numeric `age` p-value 0.0018 34 0.2694 0.1562 FALSE -1
## numeric `maturity` p-value 0.0006 33 0.3265 0.4727 <NA> 1
```

Simulation Results cont.

WoE Tables

```
##      rf      bin no  ng  nb      woe
## balance      1 273 139 134 -0.8107
## balance      2 268 163 105 -0.4075
## balance      3  63  49  14  0.4055
## balance      4 386 342  44  1.2033
## balance missing 10   7   3  0.0000
## rf.inter      1 219 160  59  0.1503
## rf.inter      2 224 189  35  0.8391
## rf.inter      3 392 264 128 -0.1234
## rf.inter      4 165  87  78 -0.7381
```

Integrated PD Model

```
##      Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -0.8486      0.0774 -10.9705      0
## balance      -1.0014      0.0979 -10.2297      0
## rf.inter      -1.0020      0.1586  -6.3194      0
```

Conclusions

- One possible use case of ML in PD modeling is to support the risk factor engineering process.
- Extracting risk factor interactions from tree-based models can provide additional benefits to the traditional PD modeling approach. Beyond the typical improvement in discriminatory power, this approach can also enhance the robustness of the final model by selecting a broader range of relevant risk factors.
- The same use case of tree-based interactions can be easily extended to Loss Given Default (LGD) and Exposure at Default (EAD) models, where practitioners often face greater challenges in achieving the required model performance.
- In addition to the presented use case, the same idea can be extended to a modular approach to model development, different designs for combining risk factors, as well as other statistical methods such as random forests.
- Practitioners should be aware that once interactions are extracted and used as inputs in a regression model, the investigation of the risk differentiation function is not complete. Further critical steps include analyzing the relationship between the individual risk factors (those used to construct the interaction) and the target variable. This analysis is essential to properly assess and validate the final risk differentiation function from a business perspective.
- A model developed using this approach can serve as a challenger model, allowing practitioners to identify relevant risk factors, their interactions, and nonlinear relationships, while also using the tree structure to improve the binning step in traditional model development.